

# Episcleral brachytherapy of uveal melanoma: role of intraoperative echographic confirmation

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Received 7 June 2016

Revised 1 August 2016

Accepted 8 August 2016

Published Online First

29 August 2016

## ABSTRACT

**Purpose** To compare the rates of tumour recurrence following episcleral brachytherapy for uveal melanoma before and after implementation of intraoperative echographic confirmation of plaque placement.

**Materials and methods** All patients with primary single ciliary body or choroidal melanoma treated with iodine-125 or ruthenium-106 plaque brachytherapy between 1 January 2004 and 30 December 2013 were included. Exclusion criteria were patients with previous radiation treatment and patients who received adjuvant transpupillary thermotherapy. Since February 2007, intraoperative echographic confirmation was initiated to ensure that the plaque was centred on the tumour base and/or all tumour margins were covered by the plaque.

**Results** 252 patients were included in the study. Local tumour control after primary brachytherapy was achieved in 242/252 (96.0%). Of the 10 patients with treatment failure, 8 patients had local recurrence and 2 patients had failure to response. With the incorporation of the intraoperative echographic confirmation for plaque positioning the treatment failure rate decreased from 9.3% (5/54 patients) to 1.5% (3/198 patients). Continuous and categorical univariable predictors of recurrence were analysed for statistical significance. The only statistically significant variable was the intraoperative echographic confirmation (HR: 0.16;  $p=0.032$ ) for recurrence within the first 24 months.

**Conclusions** Intraoperative echographic confirmation of plaque placement during episcleral brachytherapy for choroidal melanoma reduces the risk of early local recurrence.

## INTRODUCTION

The Collaborative Ocular Melanoma Study (COMS) established episcleral brachytherapy as a globe-salvaging alternative to enucleation for medium-sized choroidal melanoma.<sup>1 2</sup> The treatment failure rate following episcleral brachytherapy varies widely ranging from 0% to 27% depending upon study inclusion criteria, duration of follow-up and other variables.<sup>3</sup> Local recurrence is associated with increased morbidity because of the need for repeat brachytherapy or enucleation. Moreover, a few studies have suggested increased rate of metastasis following recurrence of the tumour.<sup>4-7</sup> Therefore, achieving high local control rate is imperative to decreasing morbidity and possibly even mortality in uveal melanoma.

Optimal positioning of the episcleral plaque with respect to the tumour margins so as to deliver planned therapeutic dose to the tumour is a prerequisite for episcleral brachytherapy. In 1988,

Williams *et al* described the use of intraoperative echographic confirmation of optimal plaque placement.<sup>8</sup> Afterwards, other centres have adopted this technique<sup>9-14</sup> and reported low local tumour recurrence rates.<sup>13</sup>

However, comparative studies of outcomes with or without use of intraoperative echographic confirmation of episcleral plaque placement have not been undertaken. Comparison of data from different centres is of limited value because of variations in radiation treatment planning software, surgical techniques and plaque designs. In this study, we report on the 10-year outcomes of episcleral brachytherapy for uveal melanoma. Furthermore, we compare the experience at our centre using the same technique for treatment planning and plaque placement before and after implementation of intraoperative echographic confirmation of plaque placement.

## MATERIALS AND METHODS

An internal review board approval was obtained. Patients treated at the Department of Ophthalmic Oncology, Cole Eye Institute at the Cleveland Clinic for a primary single ciliary body or choroid melanoma with iodine-125 or ruthenium-106 plaque brachytherapy between 1 January 2004 and 30 December 2013 were included. Exclusion criteria included: patients with previous radiation treatment and patients who received adjuvant transpupillary thermotherapy (TTT). Patient demographics and clinical information were extracted from the Cleveland Clinic electronic medical records system (EpicCare EMR, Madison, Wisconsin, USA) and paper charts. Dosimetry for the plaques was performed with plaque simulator treatment planning software (Plaque Simulator V5.3.4; Bebig GmbH, Berlin, Germany) designed to give a total prescription of 85 Gy to the tumour apex. Isotopes used were iodine-125 (Model MED3631-A/M; North American Scientific, Chatsworth, CA, USA, or Isoaid Advantage Model IAI-125A, IsoAid, Port Richey, Florida, USA) and ruthenium-106 (Models CCA, CCD and CCB, with diameters of 15.3, 17.9 and 20.2 mm, respectively; Bebig GmbH). Ruthenium-106 was used during the periods it was commercially available in the USA.

As part of the initial workup, all patients underwent a detailed ophthalmoscopic examination, photography and ultrasonographic A and B scan. Tumour size (basal dimensions and height) was assessed by ophthalmoscopic examination and by ultrasonography. Tumour localisation for plaque placement was performed by the use of



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**To cite:** Aziz HA, Al Zahrani YA, Bena J, *et al.* *Br J Ophthalmol* 2017;**101**:747–751.

transillumination or indirect ophthalmoscopy, based on tumour location (anterior or posterior to the equator). In February 2007, intraoperative echographic confirmation was initiated to ensure that the plaque was centred on the tumour base and/or all tumour margins were covered by the plaque.

All intraoperative plaque placement confirmations were performed by experienced diagnostic ophthalmic echographers. A 10 MHz posterior segment probe from ophthalmic ultrasound unit Ellex Eye Cubed V2 (Minneapolis, Minnesota, USA) or the Quantel Avisio (Bozeman, Montana) were used under sterile conditions. Using a coupling agent, dynamic transverse and longitudinal transocular scans were obtained at an optimised gain. After the radioactive episcleral had been sutured with temporary knots, scans were performed to confirm that the plaque was centred on the tumour base and/or all tumour margins were covered by the plaque. For tumours extending into ciliary body, anterior margin was not assessed by echography. Plaques that were not properly placed were repositioned, and echographic confirmation was repeated to evaluate the adjusted placement. Images were recorded and filed as part of the medical record documenting plaque position.

The patients were usually evaluated for follow-up 1 week after plaque brachytherapy, and then every 3 months throughout the first year, every 6 months throughout the second through fifth years, and annually thereafter.

Treatment failure included both failure to respond to brachytherapy and local recurrence of the tumour. Failure to respond to brachytherapy was defined as increase in the tumour thickness and/or basal diameters within the first 3 months after brachytherapy. Local recurrence was defined as regrowth of the tumour that had demonstrated initial regression or stability after treatment.

## Statistical methods

Categorical factors are described using frequencies and percentages, while continuous measures are summarised as mean and SD. Comparisons of ultrasound groups on patient characteristics were made using Pearson  $\chi^2$  tests or two-sample t-tests, as appropriate. Kaplan-Meier estimation and Cox proportional hazards models were performed to identify potential predictors of recurrence. Overall recurrence was compared, as was recurrence within the first 24 months. The analysis of recurrences

within the first 24 months was performed to remove the impact of late recurrences to compare recurrence patterns over the time period where both groups had adequate follow-up, since <40% of patients with echographic confirmation of placement had follow-up beyond 2 years. Analyses were performed using SAS software (V.9.3; Cary, North Carolina, USA).

## RESULTS

For the time period of 1 January 2004 to 30 December 2013, 353 cases of choroidal and/or ciliary body melanoma were treated at the Department of Ophthalmic Oncology, Cole Eye Institute, Cleveland Clinic Foundation. Data were extracted between 1 March 2015 to 30 June 2015. Of these, 101 patients were excluded: 65 patients with less than 1-year follow-up and 36 patients had adjuvant TTT. The remaining 252 cases of ciliary body and/or choroid melanoma were included in the study. The mean age of diagnosis was  $61.1 \pm 14.0$  years. The mean time of follow-up was  $30.9 \pm 20.1$  months. Mean tumour thickness and largest diameter were  $4.0 \pm 2.1$  and  $11.3 \pm 3.2$  mm, respectively. Iodine was the isotope of choice in 215 (85.7%) of patients, whereas ruthenium was used in the remaining 36 (14.3%) patients. In the initial 54 patients, episcleral plaque placement was verified using only traditional method of visual inspection, whereas in subsequent 198 cases, intraoperative echography was performed after visual inspection to confirm adequate placement. Summary of the patients' demographics and tumour characteristics are presented in table 1.

Overall, local tumour control after primary brachytherapy was achieved in 242/252 (96.0%) patients. Of the 10 patients with treatment failure, 8 patients had local recurrence and 2 patients had failure to response. The mean age of diagnosis, mean time of follow-up and tumour characteristics and dimensions were similar in the local recurrence group when compared with the general cohort (table 2). Of those who recurred, the median time to local recurrence was 18 months (range: 7–86 months). Of the eight recurrent tumours, six had enucleation and two had a repeat brachytherapy (with tumour control at 85 and 96 months of follow-up). None of the patients developed orbital recurrence. With the incorporation of the intraoperative echographic confirmation of plaque placement, the recurrence rate decreased from 9.3% (5/54 patients) to 1.5% (3/198 patients) (table 2).

**Table 1** Episcleral brachytherapy of uveal melanoma: role of intraoperative echographic confirmation of placement.

| Factor                           | Total (N=252)   | Pre-echographic confirmation (N=54) | Post-echographic confirmation (N=198) | p Value             |
|----------------------------------|-----------------|-------------------------------------|---------------------------------------|---------------------|
| Age (years)                      | $61.1 \pm 14.0$ | $60.9 \pm 15.1$                     | $61.1 \pm 13.7$                       | 0.94 <sup>a</sup>   |
| Months of follow-up              | $30.9 \pm 20.1$ | $50.4 \pm 28.1$                     | $25.6 \pm 13.1$                       | <0.001 <sup>a</sup> |
| Tumour height                    | $4.0 \pm 2.1$   | $3.6 \pm 1.8$                       | $4.1 \pm 2.2$                         | 0.081 <sup>a</sup>  |
| Largest basal diameter           | $11.3 \pm 3.2$  | $11.1 \pm 3.1$                      | $11.3 \pm 3.2$                        | 0.62 <sup>a</sup>   |
| Distance to fovea                | $5.2 \pm 4.6$   | $5.1 \pm 4.6$                       | $5.2 \pm 4.6$                         | 0.96 <sup>a</sup>   |
| Total dose to apex               | $86.0 \pm 9.1$  | $90.1 \pm 18.0$                     | $84.9 \pm 3.8$                        | <0.001 <sup>a</sup> |
| Sex, male                        | 121 (48.0%)     | 27 (50.0%)                          | 94 (47.5%)                            | 0.74 <sup>c</sup>   |
| Isotope, I-125                   | 215 (85.7%)     | 28 (52.8%)                          | 187 (94.4%)                           | <0.001 <sup>c</sup> |
| Distance to fovea ( $\geq 3$ mm) | 161 (63.9%)     | 35 (64.8%)                          | 126 (63.6%)                           | 0.87 <sup>c</sup>   |
| Distance to disc ( $\geq 3$ mm)  | 208 (82.5%)     | 47 (87.0%)                          | 161 (81.3%)                           | 0.33 <sup>c</sup>   |
| Recurrence (total)               | 8 (3.2%)        | 5 (9.3%)                            | 3 (1.5%)                              | 0.004 <sup>c</sup>  |
| Recurrence within 24 months      | 6 (2.4%)        | 4 (7.4%)                            | 2 (1.0%)                              | 0.020 <sup>d</sup>  |

Patient demographics and tumour characteristics.

Values are presented as mean $\pm$ SD or N (column %).

p Values: a=analysis of variance, c=Pearson's  $\chi^2$  test, d=Fisher's exact test.

**Table 2** Episcleral brachytherapy of uveal melanoma: role of intraoperative echographic confirmation of placement

|                                 | Total (N=8)  | Pre-echographic confirmation (N=5) | Post-echographic confirmation (N=3) |
|---------------------------------|--------------|------------------------------------|-------------------------------------|
| Recurrent tumours/total tumours | 8/252 (3.2%) | 5/54 (9.3%)                        | 3/198 (1.5%)                        |
| Age (years)                     | 66.4±8.5     | 68.2±6.0                           | 63.3±12.5                           |
| Months of follow-up             | 41.0±32.8    | 38.0±33.5                          | 46.0±38.2                           |
| Total dose to apex              | 83.6±3.2     | 83.5±3.9                           | 83.6±2.7                            |
| Distance to fovea               | 6.3±6.2      | 7.5±7.1                            | 4.3±4.9                             |
| Apex                            | 4.4±2.5      | 4.0±1.9                            | 5.0±3.8                             |
| Largest basal diameter          | 12.6±4.4     | 12.6±3.4                           | 12.7±6.8                            |
| Isotope                         |              |                                    |                                     |
| I-125                           | 6 (75.0%)    | 3 (60.0%)                          | 3 (100.0%)                          |
| Ru-106                          | 2 (25.0%)    | 2 (40.0%)                          | 0 (0.0%)                            |
| Sex                             |              |                                    |                                     |
| Female                          | 7 (87.5%)    | 5 (100.0%)                         | 2 (66.7%)                           |
| Male                            | 1 (12.5%)    | 0 (0.0%)                           | 1 (33.3%)                           |
| Distance to fovea (≥3 mm)       |              |                                    |                                     |
| No                              | 4 (50.0%)    | 2 (40.0%)                          | 2 (66.7%)                           |
| Yes                             | 4 (50.0%)    | 3 (60.0%)                          | 1 (33.3%)                           |
| Distance to disc (≥3 mm)        |              |                                    |                                     |
| No                              | 2 (25.0%)    | 1 (20.0%)                          | 1 (33.3%)                           |
| Yes                             | 6 (75.0%)    | 4 (80.0%)                          | 2 (66.7%)                           |
| Secondary treatment             |              |                                    |                                     |
| Enucleation                     | 6 (75%)      | 4 (80%)                            | 2 (66.7%)                           |
| Repeat brachytherapy            | 2 (25%)      | 1 (20%)                            | 1 (33.3%)                           |

Demographics and tumour characteristics of recurrent tumours.

**Table 3** Episcleral brachytherapy of uveal melanoma: role of intraoperative echographic confirmation of placement

| Parameter              | Recurrence within 24 months |         | Recurrence within 36 months |         |
|------------------------|-----------------------------|---------|-----------------------------|---------|
|                        | HR (95% CI)                 | p Value | HR (95% CI)                 | p Value |
| Age at presentation    | 1.06 (0.99 to 1.14)         | 0.091   | 1.04 (0.98 to 1.10)         | 0.16    |
| Tumour height          | 1.09 (0.76 to 1.56)         | 0.63    | 1.14 (0.83 to 1.57)         | 0.41    |
| Largest basal diameter | 1.10 (0.85 to 1.42)         | 0.46    | 1.20 (0.95 to 1.53)         | 0.13    |
| Distance to fovea      | 1.06 (0.91 to 1.25)         | 0.46    | 1.07 (0.92 to 1.24)         | 0.38    |
| Distance to optic disc | 1.01 (0.82 to 1.25)         | 0.90    | 1.05 (0.87 to 1.26)         | 0.63    |
| Total dose to apex     | 0.95 (0.86 to 1.04)         | 0.24    | 0.95 (0.87 to 1.04)         | 0.29    |

Univariable continuous predictors of recurrence within 24 and 36 months.

**Table 4** Episcleral brachytherapy of uveal melanoma: role of intraoperative echographic confirmation of placement

| Group                    | Level  | N   | Recurrence free at 24 months | Recurrence free at 36 months | HR (95% CI)         | p Value |
|--------------------------|--------|-----|------------------------------|------------------------------|---------------------|---------|
| Overall                  |        | 252 | 97.2 (95.0, 99.4)            | 97.2 (95.0, 99.4)            |                     |         |
| Echographic confirmation | No     | 54  | 92.3 (85.3, 99.8)            | 92.3 (85.3, 99.8)            | 1.00 (REF)          | 0.032   |
|                          | Yes    | 198 | 99.0 (97.6, 100.0)           | 99.0 (97.6, 100.0)           | 0.16 (0.03 to 0.85) |         |
| Isotope                  | Ru-106 | 36  | 93.5 (85.0, 100.0)           | 93.5 (85.0, 100.0)           | 1.00 (REF)          | 0.22    |
|                          | I-125  | 215 | 97.9 (95.9, 100.0)           | 97.9 (95.9, 100.0)           | 0.34 (0.06 to 1.88) |         |

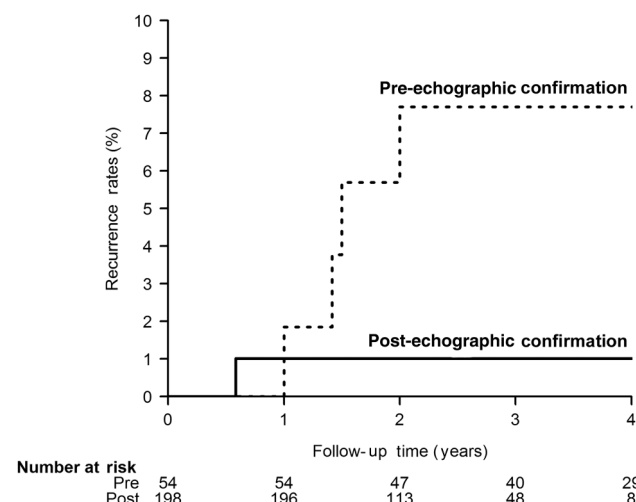
Univariable categorical predictors of recurrence within 24 and 36 months.

Continuous and categorical univariable predictors of recurrence were analysed for statistical significance within the first 24 months and throughout the duration of the study. Continuous univariable factors included: age at presentation, tumour dimensions, and location and total radiation dose delivered to the apex. None of those parameters reached statistical significance (table 3). Categorical univariable factors included: use of intraoperative echographic confirmation of plaque positioning and the type of isotope (table 4). The only statistically

significant variable was the intraoperative echographic confirmation (HR: 0.16;  $p=0.032$ ) when analysed for recurrence within the first 24 months (figure 1).

## DISCUSSION

In the current study, the local control rate for choroidal melanoma after instituting intraoperative echographic confirmation for plaque positioning is 98.5%. This is consistent with the excellent local control rates reported by different centres using



**Figure 1** Kaplan-Meier estimates of local recurrence and the log-rank test between cohorts. Follow-up is shown through 4 years, where 10% of patients remain active. One additional late recurrence occurred in each group after 4 years (not shown in the plot).

echographic confirmation of plaque placement.<sup>12–14</sup> The weighted recurrence rate of the 1011 patients reported in these centres was 1.7% compared with the 10.3% reported by the COMS study<sup>15</sup> (table 5). These data strongly indicate that the use of intraoperative echographic confirmation for plaque placement helps in reducing early local recurrence in choroidal melanoma. As satisfactory plaque placement is associated with learning curve, intraoperative echographic confirmation will be of particular benefit to less-experienced surgeons.<sup>16</sup>

In a recent review paper, Chang and McCannel<sup>3</sup> discussed the local treatment failure after globe-conserving therapy for choroidal melanoma. They compared the outcomes of centres that use intraoperative echographic confirmation versus traditional method of visual inspection and suggested that centres using the former method have low recurrence rates. However, variations in radiation treatment planning software, surgical techniques and plaque designs between centres limit direct comparison of results.

In the current study, we compare the local recurrence rates with the same surgical technique before and after the introduction of intraoperative echographic confirmation of plaque placement. The distribution of the tumours (proximity to disc or fovea) was similar between the pre-echographic and post-echographic cohorts (table 1). Furthermore, to eliminate the

impact of adjuvant therapy such as TTT in juxtapapillary tumours, those patients were excluded from the study. However, there were some differences between the pre-echographic and post-echographic cohorts that include: (a) type of isotope used where ruthenium-106 was used predominantly in the pre-echographic cohort and (b) a slightly higher total dose was delivered to tumour's apex in the pre-echographic cohort that was attributed to the use of ruthenium-106. This was not found to be a statistically significant risk factor for recurrence within 24 or 36 months (tables 1 and 3).

The overall local recurrence rate dropped from 9.3% to 1.5% after instituting intraoperative echographic confirmation as part of the surgical protocol. To account for lead-time bias secondary to a longer follow-up in the pre-echographic cohort, statistical analysis of the local recurrence rate within the first 24 months was calculated. The results showed a significantly lower recurrence rate after the introduction of intraoperative echographic confirmation. Moreover, the proximity to disc or fovea was not found to be a statistically significant risk factor for recurrence within 24 or 36 months (table 3).

The limitations of our study include the retrospective collection of clinical data, the relatively small number of positive events (local recurrences) and the limited follow-up data that could potentially exclude cases of late recurrences (>24 months post-treatment). Moreover, other potential factors that differed between time periods that cannot be measured include the surgeon's experience. On the other hand, the strengths of the study include the use of the same surgical placement technique and the same continued and uniform follow-up pattern on most of the patients. Additionally, the use of a standard radiation treatment planning protocol with validated software provided reproducible data to compare the treatment failure rates before and after implementation of intraoperative echographic confirmation for plaque positioning.

In conclusion, episcleral brachytherapy offers excellent local control rates for selected cases of uveal melanoma. Furthermore, the use of intraoperative echographic confirmation of plaque placement is a simple technique that may decrease the rate of early local recurrences, thus decreasing the morbidity of secondary treatment and perhaps, even tumour-specific mortality. Given the relatively small numbers of recurrences with episcleral brachytherapy, drawing statistical significance is challenging. Therefore, a multicentre prospective study is needed to validate our observations.

**Contributors** HAA, YAAZ, BL and JB had substantial contributions to the conception or design of the work, or the acquisition, analysis or interpretation of data. ADS, AW and JS were responsible for critically revising the important intellectual content of the manuscript. ADS and HAA gave the final approval of the version published. All authors agree to be accountable for all aspects of the work in

**Table 5** Episcleral brachytherapy of uveal melanoma: recurrence rates with intraoperative echographic confirmation of plaque placement<sup>12–15</sup>

|                                                                         | Isotope                      | Number of patients | Local recurrence rate (%) | Mean largest basal diameter (mm) | Mean tumour height (mm) | Mean follow-up (months) |
|-------------------------------------------------------------------------|------------------------------|--------------------|---------------------------|----------------------------------|-------------------------|-------------------------|
| <i>COMS: echographic confirmation was not used for plaque placement</i> |                              |                    |                           |                                  |                         |                         |
| Jampol et al <sup>15</sup>                                              | Iodine-125                   | 650                | 10.3                      | 11.5                             | 4.2                     | 60                      |
| <i>Studies that used echographic confirmation for plaque placement</i>  |                              |                    |                           |                                  |                         |                         |
| Tabandeh et al <sup>12</sup>                                            | Iodine-125                   | 117                | 1.7                       | 9.6                              | 4.2                     | 37                      |
| McCannel et al <sup>13</sup>                                            | Iodine-125                   | 170                | 0                         | 10.8                             | 4.8                     | 32.4                    |
| Badiyan et al <sup>14</sup>                                             | Iodine-125                   | 526                | 3.6                       | 12.2                             | 4.2                     | 53.4                    |
| Current study                                                           | Iodine-125 and ruthenium-106 | 198                | 1.5                       | 11.3                             | 4.1                     | 25.6                    |
| Total                                                                   |                              | 1011               | 1.7 (weighted average)    |                                  |                         |                         |

ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

**Competing interests** None declared.

**Ethics approval** Institutional Review Board.

**Provenance and peer review** Not commissioned; externally peer reviewed.

## REFERENCES

- Melia BM, Abramson DH, Albert DM, *et al.* Collaborative ocular melanoma study (COMS) randomized trial of I-125 brachytherapy for medium choroidal melanoma. I. Visual acuity after 3 years COMS report no. 16. *Ophthalmology* 2001;108:348–66.
- Krintz AL, Hanson WF, Ibbott GS, *et al.* A reanalysis of the collaborative ocular melanoma study medium tumor trial eye plaque dosimetry. *Int J Radiat Oncol Biol Phys* 2003;56:889–98.
- Chang MY, McCannel TA. Local treatment failure after globe-conserving therapy for choroidal melanoma. *Br J Ophthalmol* 2013;97:804–11.
- Vrabec TR, Augsburger JJ, Gamel JW, *et al.* Impact of local tumor relapse on patient survival after cobalt 60 plaque radiotherapy. *Ophthalmology* 1991;98:984–8.
- Char DH, Kroll S, Phillips TL. Uveal melanoma. Growth rate and prognosis. *Arch Ophthalmol* 1997;115:1014–18.
- Harbour JW, Char DH, Kroll S, *et al.* Metastatic risk for distinct patterns of postirradiation local recurrence of posterior uveal melanoma. *Ophthalmology* 1997;104:1785–92.
- Ophthalmic Oncology Task Force. Local Recurrence Significantly Increases the Risk of Metastatic Uveal Melanoma. *Ophthalmology* 2016;123:86–91.
- Williams DF, Mieler WF, Lewandowski M, *et al.* Echographic verification of radioactive plaque position in the treatment of melanomas. *Arch Ophthalmol* 1988;106:1623–4.
- Harbour JW, Murray TG, Byrne SF, *et al.* Intraoperative echographic localization of iodine 125 episcleral radioactive plaques for posterior uveal melanoma. *Retina (Philadelphia, Pa)* 1996;16:129–34.
- Finger PT, Romero JM, Rosen RB, *et al.* Three-dimensional ultrasonography of choroidal melanoma: localization of radioactive eye plaques. *Arch Ophthalmol* 1998;116:305–12.
- Al Mahmoud T, Mansour M, Deschênes J, *et al.* Intra-operative echographic localization for radioactive ophthalmic plaques in choroidal melanoma. *Ann N Y Acad Sci* 2008;1138:10–14.
- Tabandeh H, Chaudhry NA, Murray TG, *et al.* Intraoperative echographic localization of iodine-125 episcleral plaque for brachytherapy of choroidal melanoma. *Am J Ophthalmol* 2000;129:199–204.
- McCannel TA, Chang MY, Burgess BL. Multi-year follow-up of fine-needle aspiration biopsy in choroidal melanoma. *Ophthalmology* 2012;119:606–10.
- Badiyan SN, Rao RC, Apicelli AJ, *et al.* Outcomes of iodine-125 plaque brachytherapy for uveal melanoma with intraoperative ultrasonography and supplemental transpupillary thermotherapy. *Int J Radiat Oncol Biol Phys* 2014;88:801–5.
- Jampol LM, Moy CS, Murray TG, *et al.* Collaborative Ocular Melanoma Study Group (COMS Group). The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: IV. Local treatment failure and enucleation in the first 5 years after brachytherapy. COMS report no. 19. *Ophthalmology* 2002;109:2197–206.
- Shah NV, Houston SK, Murray TG, *et al.* Evaluation of the surgical learning curve for I-125 episcleral plaque placement for the treatment of posterior uveal melanoma: a two decade review. *Clin Ophthalmol* 2012;6:447–52.